SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Yaponomics
Project / Solution Name	Yapoz
Name (Leader)	Abdulrahman Imaad Mohamed
Email & Phone (Leader)	swe2409099@xmu.edu.my, +60173793811
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://yapoz.vercel.app (judges: use the demo buttons on the login page)
Code Repository URL (public / access granted to judges)	https://github.com/Ramilbey/yapoz

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface to backend to AI model/API to database to cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Customers contact a business through WhatsApp, Telegram or the website simulator. Messaging webhooks verify Meta signatures or Telegram secret tokens; the simulator uses its own endpoint. The Next.js app runs on Vercel, with PostgreSQL on Supabase storing messages, jobs and business records. Failed jobs can retry. The shared pipeline loads catalogue, hours, policies and FAQs. The AI router tries the enabled KrackedDevs gateway first, with Groq and Google Gemini as fallbacks. Tools retrieve business facts; code calculates prices, checks stock and saves pending orders. AI replies can still be inaccurate. A vision model reads receipt photos, and rules compare the extracted details with the order. An owner or service user reviews and approves or rejects. Backend checks enforce role, status, stock and receipt requirements. Customer notifications and activity events record the outcome. PostgreSQL LISTEN/NOTIFY supports live updates. Cloudflare Turnstile and Resend support sign-up verification. Four Orbyte event formats and signing logic are implemented, but no events are sent to Orbyte.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 7.0.2	Language for the whole codebase	Apache-2.0
Next.js 16.4.2	Web app, API routes, server rendering	MIT
React and React DOM 19.3.1	User interface	MIT
Tailwind CSS 4.3.3	Styling	MIT

Name & Version	Purpose in Project	License
Radix UI: dialog 1.3.1, dropdown-menu 2.2.1, popover 1.3.1, switch 1.4.1, tabs 1.2.1, tooltip 1.4.1	Accessible interface components	MIT
SWR 2.5.1	Client data fetching and live refresh	MIT
Drizzle ORM 0.45.4, Drizzle Kit 0.31.11	Database access and migrations	Apache-2.0, MIT
postgres (postgres.js) 3.4.9	PostgreSQL driver	Unlicense
Zod 4.6.7	Validation of every API request	MIT
jose 6.2.13, bcryptjs 3.0.3	Signed sessions, password hashing	MIT, BSD-3-Clause
grammY 1.46.0	Telegram bot framework	MIT
OpenAI SDK 7.39.0	Client for OpenAI-compatible AI APIs (KrackedDevs gateway, Groq, Gemini)	Apache-2.0
Anthropic SDK 0.135.0	Optional Claude adapter, not enabled in production	Apache-2.0
web-push 3.6.7	Browser push alerts to staff	MPL-2.0
unpdf 1.9.1, pdf-lib 1.17.1, qrcode 1.5.4	Read PDFs, generate sample PDFs, QR codes for customer links	MIT
date-fns 4.4.0, @date-fns/tz 1.5.0, city-timezones 1.3.4	Dates, time zones and place lookup	MIT
clsx 2.1.1, server-only 0.0.1	Class names, server-only module guard	MIT
Three.js 0.186.1, React Three Fiber 9.8.2, drei 10.7.9, postprocessing 6.39.5	3D landing page	MIT (postprocessing: Zlib)
GSAP 3.15.0, @gsap/react 2.1.2, Motion 14.4.0, Lenis 1.3.27	Animation and smooth scrolling	GSAP Standard "no charge" licence; MIT
Fontsource: Bricolage Grotesque, JetBrains Mono, Instrument Serif	Web fonts	OFL-1.1

Name & Version	Purpose in Project	License
tsx 4.23.15, ESLint 10.13.0, Playwright 1.64.1, fflate 0.8.3	Development tooling: scripts, linting, screenshots	MIT, Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
KrackedDevs LiteLLM gateway (litellm.rapidscreen.io)	AI replies, message classification and tool calling. First choice when its key is set; the other providers are the fallback	sponsor-provided
Groq API (Groq)	AI replies, classification and tool calling (gpt-oss models; Llama and Qwen as fallback)	free
Gemini API (Google)	Reading receipt and menu photos (vision); chat fallback	free
WhatsApp Cloud API (Meta)	Receive customer messages, send replies, download receipt files (Meta test number)	free
Telegram Bot API (Telegram)	Receive customer messages, send replies, download receipt files	free
Google Calendar API (Google)	One-way copy of approved bookings to a business's calendar; off unless the business connects it	free
Cloudflare Turnstile (Cloudflare)	Human check at sign-up	free
Resend (Resend)	Sends the 6-digit sign-up code by email	free
Web Push (browser push services)	Alerts to staff browsers that have turned them on	free
Anthropic, OpenAI, OpenRouter and Ollama APIs	Supported by the AI router but disabled in production: no keys are set	not used

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel (Vercel Inc.)	Hosts the web app and API as serverless functions and runs the daily retry sweep. Deployed from the GitHub main branch	Hobby (free), functions in Washington D.C. (iad1)
Supabase (PostgreSQL)	Database, job queue, live dashboard updates	Free tier, Seoul (aws-0-ap-northeast-2)
GitHub	Public source code hosting; pushes to main trigger the Vercel deployment	Free
Cloudflare (Turnstile)	Sign-up human check	Free
Resend	Email delivery for sign-up codes	Free
Replit	Used during development for the initial project structure; the app is not deployed there	Sponsor-provided, not a deployment region

Additional external service (Sections 3.1-3.2)
Jitsi Meet (meet.jit.si): public video-meeting rooms for online appointments. Yapoz generates a room link; participants connect to Jitsi when they join. Free public service; hosting region is not controlled by the team. Jitsi/provider terms apply. This is separate from the optional Google Calendar connection.

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-oss-120b and gpt-oss-20b	OpenAI, served by Groq	pre-trained, used as is	Apache-2.0
Llama and Qwen models (fallback only; the version is chosen automatically from Groq's list)	Meta and the Qwen team (Alibaba), served by Groq	pre-trained, used as is	As published on Groq's model pages
Gemini Flash and Flash-Lite (newest stable version, chosen automatically)	Google, Gemini API	pre-trained, used as is	Google Gemini API terms
gpt-5.4-nano gpt-5.4-mini	OpenAI models via KrackedDevs gateway	pre-trained, used as is	Gateway and vendor terms; retention and training terms not stated to us

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Forno Sette (FORNO7), Kasih Hair Studio (KASIH1) and Dapur Mak Long Catering (CATER1): menus, prices, hours, FAQs, bookings, orders and demo conversations	Created by the team (seed scripts and data in the repository)	synthetic	Created by the team (Apache-2.0 with the repository)	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic orders and sample receipts for the 20-order measurement run	Generated by the team's script (scripts/synthetic-orders.ts)	synthetic	Created by the team	N
Test conversations and test sign-ups on the live deployment	Messages sent by team members and testers	private, self-collected	Not shared	Y (display names, chat IDs, a WhatsApp number, email addresses)

Model configuration detail (Section 4.1)

The team confirms that the live KrackedDevs gateway uses gpt-5.4-nano for fast turns and gpt-5.4-mini for harder turns.

Groq code defaults: openai/gpt-oss-20b and openai/gpt-oss-120b. Named fallback candidates include llama-3.3-70b-versatile, llama-3.1-8b-instant and qwen/qwen3-32b. The discovery filter also permits available Kimi models from Moonshot AI. These are supported candidates, not a claim that each ran in the demo.

Gemini code defaults: gemini-3.5-flash-lite and gemini-3.8-flash. Unless pinned through configuration, the router discovers available Flash/Flash-Lite models, so actual selection can vary. Activity records identify models used for individual replies. Provider and model-specific terms apply.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic), with Claude Sonnet 5.5 and Opus 5.5	Writing and testing code and database migrations, debugging, and drafting the hackathon documents	major
Replit Agent	Initial project structure	moderate
OpenAI Codex	Coding, development and debugging assistance	moderate
ChatGPT (OpenAI)	Pitch script, presentation planning, documentation and README assistance	minor
Claude (Anthropic)	Slide-deck generation and presentation assistance	minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team reviews AI-assisted changes using TypeScript checks, production builds, automated tests on a disposable database and manual customer-chat tests. The repository contains over 250 tests; a test count alone is not proof that every deployment passed. Backend code computes catalogue prices, enforces approval and records activity. Receipt findings remain advisory, and staff review consequential actions. Tests cover approval, tenant isolation, provider failures and injection attempts. The team reviews generated text and code, corrects observed errors, and takes responsibility for the submitted work, licences and security.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
KrackedDevs (via the SDC Hackathon)	AI API credit, US$100 per team, as personal OpenAI-compatible keys	The AI router calls the KrackedDevs gateway first for replies and tool calling. Groq and Gemini answer if it fails or the credit ends (7 Nov 2026)
Replit (via the SDC Hackathon)	Replit Core plan and Agent credits	Replit Agent produced the initial project structure. The credits ran out, so development continued with Claude Code
Runpod (via the SDC Hackathon)	Runpod credit, US$25 per team	Not used: the team neither trains nor hosts a model

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No. KrackedDevs is tried first, with configured Groq and Gemini providers as fallbacks. Sponsor credit ends on 7 November 2026. Fallback success depends on provider availability and quotas; if all fail, staff intervention is needed. Replit supported initial development but does not host the app. Runpod was not used because the team neither trains nor hosts models.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Yapoz stores customer display names, chat IDs or WhatsApp numbers, messages, attachments, orders and bookings. Staff accounts store names, emails and password hashes. Messages go to configured AI providers (KrackedDevs, Groq and Google Gemini); images go to vision-capable providers. The gateway forwards requests to model vendors; its retention and training terms have not been confirmed. We do not train models on this data. Stored channel credentials use AES-256-GCM; passwords use bcrypt. Workspace and role checks restrict access, and logging redacts secrets. Settings offer 30-365 day retention; eligible conversations are purged and finished orders anonymised. Retention is off by default. Demo business records are synthetic; tester account and channel identifiers may be personal data. Joining a Jitsi room also sends participants' connection and meeting data to that service under its terms.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

AI replies and receipt extraction can be inaccurate. Catalogue tools, code-calculated prices, FAQ retrieval and human escalation reduce risk but do not eliminate hallucinations. Receipts can be forged or misread; consistency checks do not verify bank payment. Staff approve orders, and receipt text is treated as untrusted data, with injection tests. Gateway credits expire on 7 November 2026; provider quotas, outages and model changes can interrupt replies. Routing, retries and staff alerts mitigate these failures but cannot guarantee availability. The current WhatsApp reply flow depends on the customer-service window; approved template messaging outside that window is not implemented. No real business owner has validated the product. Retention requires owner configuration. Orbyte payloads are previews only, with no live integration.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

☒ Any use of official SDC sponsor resources has been declared in Section 5.

☒ The prototype described here is accessible to judges through the links provided in Section 1.

☒ All fields in this template have been completed within the word limits stated.

Credential clarification: README lists intentionally public demo logins for synthetic workspaces. These are test credentials; production API keys and private account credentials must remain secret.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



Abdulrahman Imaad Mohamed | Team Leader

Date: 1 November 2026